Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/02/2004
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Project Title:	Biosensors for real-time monitoring of radiation induced biologic effects in space.		
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Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
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Zip Code:	48109-0648	**Congressional District:**	15
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	We propose to develop nanoscale biosensors and bioactuators for use in astronaut health and safety monitoring, in conjunction with external analytic bioNEMS devices. This would involve the evolution of nanoscale polymer structures less than 20 nm in diameter as the basis of the sensor/actuators. The structures would be designed to target into specific cells of an astronaut and be able to monitor health issues such as the exposure to radiation or infectious agents. These molecules would also be able to administer therapeutics in response to the needs of the astronaut, and act as actuators to remotely manipulate an astronaut as necessary to ensure their safety. An example of this latter capability would be to activate muscle movement causing an unconscious astronaut to walk back into a spacecraft and out of harm's way. In addition, the nanosensor could be used to develop cellular-based biochips that would be useful as sensors for the remote detection of life or other scientific analysis in space. These studies will be truly multidisciplinary, involving a fusion of disciplines including nanotechnology-based materials science, bioengineering, bioinformatics and medical sciences. We will use these different disciplines to converge on the		

Task Description:

design and manipulation of the nanosensors, and the development of non-invasive systems to interact with the sensors through functional MRI or multispectral fluorescence imaging. This latter task would require the engineering of wearable NEMS systems by engineers involved in this project while other members of the project team would test the efficacy of these systems using biologic models, such as experimental animals. Because of these broad requirements, the research would involve the multidisciplinary team from the Medical, Engineering and basic science (LS&A) schools at the University of Michigan, and would train multidisciplinary scientists at the pre-and postgraduate level. We will partner with a historically black institution, Howard University, for supplemental materials science capability necessary for this program. This team would develop strong interactions with NASA scientists such as the Nanotechnology group at the Ames research center and multispectral imaging scientists, and allow NASA scientists opportunities for training at the lead institution. Objective I: The development of dendritic polymer based biosensors capable of monitoring radiation-induced changes in lymphocytic populations. Objective II: The analysis and verification of the radiation sensing capability in lymphocytes in vitro. Objective III: The development of a noninvasive laser analysis system to monitor biosensor signals of radiation-induced damage in lymphocytes. Exposure to ionizing radiation is an important problem in manned space travel. Exposure to radiation can cause both acute and chronic illness, and the spacecraft itself provides little shielding to the radiation that could be encountered in space. Acute, high-level radiation exposure of greater than 2500 cGy will inactivate lymphocytes and induce apoptosis in the majority of these cells, leaving the individual susceptible to infectious diseases (American Journal of Industrial Medicine 20:805-10, 1991, Stem Cells 2:275-85, 1997). Higher doses of radiation can prevent the replication of cells in the respiratory and gastrointestinal mucosa as well as platelets and granulocytes (Military Medicine 165:275-7, Progress in Clinical & Biological Research. 372:155-65, 1991, British Journal of Hospital Medicine 39:498-504, 1988). This leaves the subject extremely ill and death usually occurs due to bone marrow failure or hemorrhage from a combination of inadequate platelet function and mucosal breakdown. Low-level radiation exposure can also cause illness, in a more protracted manner with serious heath compromise. This type of radiation exposure can cause genetic damage that can predispose to a number of neoplasms, particularly lymphomas (Health Physics 56:821-38, 1989, Medicine, Conflict & Survival 14:106-19, 1998). Lymphocytes are sensitive to radiation effects and therefore are likely to have radiation-induced damage that can cause neoplasms (Eisenberg RL. Radiation injury and protection. In: Radiology: an illustrated history. St. Louis: Mosby Year Book; p. 156-82, 1992). However, with prolonged exposure to low doses of radiation many types of cancers can be observed (Radiation Research 144:160-9, 1995).

While it is medically important to monitor real-time radiation exposure in individuals, no techniques are currently suitable to do this. There are radiation monitoring devices in spacecraft, but these do not measure actual dosing to the astronauts or anticipate the biologic consequences of exposure. In addition, it is difficult to calculate the radiation dose that an astronaut might accumulate in the multiple environments he or she encounters in space (either in or outside of the spacecraft). Individuals also may have different susceptibilities to radiation-induced illness, making the measure of a specific “biological dose” for each individual imperative (International Journal of Radiation Oncology, Biology and Physics 30:997-1002, 1994, International Journal of Radiation Oncology, Biology and Physics 42:379-89, 2001). There are, however, techniques to measure the “biologic dose” of ionizing radiation and these center around the sensitivity of lymphocytes to radiation exposure. This sensitivity allows acute radiation exposure to be identified by first identifying the percentage of lymphocytes undergoing apoptosis in peripheral blood and subsequently through monitoring lymphocyte counts for the gradual disappearance of the cells after undergoing cell death (Stem Cells 15:S269-74, 1997). Chronic radiation exposure at low-levels can also be evaluated in lymphocytes through the monitoring of cytogenetic changes (Radiation Research 140:134-42, 1994, Mutation Res 339:177-89, 1995, Radiation Research 146:479-82, 1996, Radiation Research 148:S33-8, 1997), although this is somewhat controversial (Radiation Research 147:506-10, 1997). A sensitive marker for chronic exposure is the frequency of reciprocal translocations in the chromosomes of blood lymphocytes which can be identified through fluorescence in situ hybridization (Radiation Research 33:S206-14, 1992). This is a consistent marker that is not affected by other metabolic or environmental toxins but requires that cells must be removed from the individual and analyzed either with microscopy or some type of analytic machine, such as a flow cytometer (Medical Physics 7:609-15, 1980). However, neither of these techniques can be performed in space because of the equipment required, the necessity for special processing of the cells plus the limitations of a zero gravity environment. Even if a device could be developed that would allow for the analysis of blood samples in zero gravity, the necessity to draw blood would still create technical difficulties. Importantly, this testing cannot be done continuously to provide real-time analysis of an individual’s radiation exposure in a manner that might allow for therapeutic intervention (such as transfusion of stored, shielded autologous blood or bone marrow) (N Engl J Med 321:205-12,1989).

Optimal Solutions for Radiation Monitoring in Space It would be optimal to monitor the biologic effects of radiation during space travel using a system that could measure exposure in a real-time, continuous manner. This should be done non-invasively as to avoid multiple sampling or blood drawing and therefore could provide momentary information on the astronaut’s status. By necessity, this monitoring approach would require some type of highly miniaturized device that could assess radiation effects in target cells, such as circulating lymphocytes, in a reliable and specific manner. It would also allow collecting data that would measure and differentiate acute as well as chronic exposure. A valuable addition would be that this system could measure the effects of other physiologic perimeters non-invasively. These applications might include calcium mobilization, as a measure of bone demineralization from zero gravity, or screening neutrophils for evidence of infection. This latter point is important since it became clear during the Mir mission that spacecraft do not have sterile environments, particularly after long-term inhabitation. Any system that would monitor these parameters non-invasively would also have to be able to operate in zero gravity space environment with low-power requirements and also with minimal weight and size to make it practical for inclusion on a long-term space voyage. These additional requirements emphasize the potential benefit of a monitoring system that could be adaptable to multiple monitoring applications avoiding the need for redundant technology.

We believe that the development of a “biologic flow cytometer” for the non-invasive monitoring of radiation-induced damage to lymphocytes would be an exciting and highly valuable goal to accomplish and could fulfill all the requirements of a non-invasive radiation monitoring system. We have developed similar types of nanosensors for other applications, documented that they insert and function within cells and can provide multi-spectral signals that can be measured on a real-time basis. Importantly, our colleagues have knowledge of a non-invasive laser system for monitoring that is small enough to even be wearable to provide continuous data from astronauts. Thus, based on preliminary work in other disease systems (see below), we have confidence that in vitro studies developing lymphocyte targeting biosensors and a demonstration of the concept of a laser-monitoring device can be accomplished within the three-year timeframe of the proposal.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

I. Executive Summary This work seeks to develop cellular biosensors based on dendritic polymers, nanoscale structures less than 20 nm in diameter. These sensors will be used to target into specific cells of an astronaut and be able to monitor health issues such as exposure to radiation. Multiple chemical structures can be assembled on the polymers including target directors, analytical devices (such as molecular probes), and reporting agents. Reporting will be accomplished through fluorescence signal monitoring with the use of multispectral analysis for signal interpretation. These nanosensors could facilitate the success and increase the safety of extended space flight. The design and assembly of these devices has been pioneered at the Center for Biologic Nanotechnology in the University of Michigan.
Scope of work: We have developed two lymphocyte-targeting nanodevices with three alternative fluorescent probes that can measure parameters of radiation-induced damage. One involves the assessment of an early marker, mitochondrial membrane disruption, while the others monitor a late marker, activated caspase 3 activity. We have also developed photonic systems capable of identifying fluorescent signals from individual cells in vivo.

Expected deliverable: This work exceeds our expected deliverable of at least one nanodevice biosensor that can target lymphocytes and measure more than one parameter of radiation-induced injury.

Decision point: Since multiple functionality, and a lymphocyte targeting nanodevice sensor were developed, the project will proceed to year 3.

Technical Objective I: Development of dendritic polymer based biosensors capable of monitoring radiation-induced changes in lymphocyte populations. Synthesis of Biosensor. Balazs Keszler, Ph.D., Istvan Majoros, Ph.D. In the first quarter of the last year, we finalized the selection of the most suitable mitochondrion dyes—MitoTracker® Deep Red (DR) and MitoTracker® Red CMXRos (CR)—for the biosensor under development. This decision was based on our synthetic work and our spectroscopic studies on the dyes in test biosensors.

The structural characteristics of G5 dendrimer-MitoTracker® Deep Red 633 (G5-DR) conjugates were verified by UV-visible spectral analysis. The test materials performed as designed with the targeted device, G5-DR-FA, associating with KB cells in a saturable fashion, and the non-targeted device, G5-DR, completely failing to associate with KB cells.

In the second quarter of the last year, we developed a lymphocyte-targeting nanodevice (a biosensor) that can measure two parameters of radiation-induced damage: (1) an early marker—mitochondrial potential changes, and (2) a late marker—caspase activity. The synthesis of the early marker detector has progressed through biological testing as reported. We have pursued two synthetic schemes in parallel for the construction of the late marker detector. These designs both use a single dye molecule that is inactive until an internal bond is cleaved by activated caspase. We have completed the first two synthetic steps in the construction of both.

In the third quarter of the last year, we synthesized a G5 dendrimer based device with one "trigger" dye, which gives a continuous fluorescence signal, and with one "reporter" dye which monitors mitochondrial activity and responds to the two photon flow cytometry system being developed under this contract.

In the fourth quarter of the last year, we completed the synthesis of the late apoptosis detector, which is based on the rhodamine-110 dye. We have attached it to both a mono-functional biosensor and a tri-functional biosensor. We have demonstrated the success of the conjugation by UV spectroscopy.

Technical Objective II: The analysis and verification of the radiation sensing capability in lymphocytes in vitro. The Use of Mouse Leucocytes to Monitor Cells In vitro and In vivo. Andrzej Myc, Ph.D., Jolanta Kukowska-Latallo, Ph.D., Jeffrey J. Landers, B.S. In the current annual reporting period, we completed in vitro testing of biocompatible dyes that do not affect viability of cells for a prolonged period of time. We were able to detect freshly isolated, live, peripheral, mononuclear, blood cells (lymphocytes from mice) stained with these dyes. We also characterized two different detection systems, the optical-fiber flow and the free-space flow, using a single dye, two dyes, or a single two-color dye.

Cell Evaluation Using Microscopic Multispectral Analysis. Felix de la Iglesia, M.D. and Timothy Sassanella, M.S. This year we: completed the uptake study of the MitoTracker® dyes and their dendrimer conjugates, showed that the uptake of these dendrimer-dye-folic acid conjugates can be inhibited in an uptake assay by excess folate, assessed the photostability and the levels of photobleaching of our test materials in vivo, and showed that the dendrimer-dye-folic acid conjugates are functioning as biosensors of mitochondrial transmembrane potential in vivo. We have also made further refinements to our analytical process.

We completed the development of the primary human lymphocyte in vitro test system. We have initiated uptake and oxidative phosphorylation decoupling experiments using this system. We are studying the MitoTracker® dyes in this system. All previously reported experiments will be repeated in this human lymphocyte cell system (many have already been finished).

Analysis of Biosensor Lajos Balogh, Ph.D. and Mohammad Islam, Ph.D. The main objective of this work is to develop cellular biosensors based on dendritic polymers. For proper functioning of these biosensors it is vital to perform extensive characterization of dendritic polymers, biosensors and intermediate products during the synthesis of the biosensors.

To achieve the objectives, molecular characterization of G5 PAMAM dendrimers with various surface functionalized groups and bi and trifunctional dendritic nanodevices were performed. High performance liquid chromatography (HPLC) and gel permeation chromatography (GPC) techniques were developed. GPC provides information primarily about the molar mass moments and hydrodynamic radius. HPLC helps to determine purity of the products based on surface characteristics. Efforts are underway to completely characterize multifunctional nanodevices and biosensors to determine their molar mass, size, purity and exact composition.

In vitro Binding Studies of Biosensor. Thommey Thomas, Ph.D. and Alina Kotlyar, M.S. We demonstrated that the G5-FITC-FA biosensor can target mouse lymphocytes—the L1210 cell line. This is proof-of-principle that we can target lymphocytes. We confirmed that the spectral and binding properties of the mitochondrial dye MitoTracker® Deep Red 633 were such that it is suitable for use in the radiation detection biosensor.

	Technical Objective III: The development of a noninvasive laser analysis system to monitor biosensor signals from lymphocytes with radiation-induced damage. Flow Cytometer-Laser System Development. Theodore Norris, Ph.D., Jingyong Ye, Ph.D., Cheng Frank Zhong, B.S. In this annual reporting period, we have demonstrated the possibility of conducting in vivo flow cytometry on a proof-of-principle T3 system, which utilizes two-photon excitation, two-beam scanning, and two-channel detection. Using the T3 flow cytometry setup, we conducted in vitro experiments to demonstrate its capability of detecting cell aggregation and quantifying viability changes in cells.
Bibliography Type:	Description: (Last Updated: 08/01/2019)
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